

M Tech (AIML)_Machine Learning _Assignment 1

Problem Statement 4

Modern healthcare screening systems often rely on **binary symptom indicators** rather than continuous clinical measurements. Such data presents unique challenges in modeling due to:

- Lack of numerical gradients
- High feature interdependence
- Interpretability requirements

You are provided with a **symptom-based patient dataset**, where each record corresponds to an individual evaluated for a potential metabolic disorder.

Your objective is to design a **machine learning pipeline that is not only accurate but clinically interpretable and robust**.

Q1. Data Understanding & Audit

a. Load the dataset and perform a **schema validation check**:

- Data types
- Unique value distributions
- Class balance

b. Identify:

- Whether features are **ordinal, nominal, or binary**
- Any inconsistencies in encoding

c. Write a short note:

Why is this dataset **not suitable for naive linear assumptions**?

Q2. Exploratory Analysis with Insight Generation

a. Perform **multi-level EDA**:

- Univariate and bivariate visualizations
- Target-wise feature behavior

b. Identify at least **3 meaningful symptom interactions** and support with visual evidence

c. Apply:

- Correlation OR mutual information

d. Critically answer:

Why correlation alone is insufficient for this dataset? Suggest a better alternative.

Q3. Preprocessing, Feature Engineering & Selection

a. Preprocess the dataset:

- Handle missing values (if any)
- Encode categorical variables (compare at least 2 methods)

b. Engineer at least **2 new features**, such as:

- Symptom severity index
- Aggregated risk score

c. Perform feature selection using:

- One filter method
- One model-based method

d. Justify:

Which features are most informative and why?

Q4. Model Development & Comparative Study

a. Train at least **3 models**:

- Logistic Regression
- Decision Tree
- One advanced model (Random Forest / SVM / Boosting)

b. Perform:

- Hyperparameter tuning
- Cross-validation

c. Compare models using:

- Accuracy, Precision, Recall, F1, ROC-AUC

d. Critically analyze:

Trade-off between interpretability and performance in this problem

Q5. Robustness, Interpretation & Deployment Thinking

a. Perform **feature importance analysis**:

- Identify top contributing symptoms

b. Conduct **robustness test**:

- Remove top features and retrain
- Observe performance impact

c. Answer:

Which features are indispensable for prediction?

d. Briefly discuss:

How would you deploy this model in a real-world clinical screening system?

Submission Guidelines

- **Submit a Jupyter Notebook (.ipynb)**
- **Include:**
 - **Code**
 - **Visualizations**
 - **Clear explanations**

Make sure to add Group members contribution at the top